

DEEP LEARNING PROJECT

CSE-A S5

GROUP- A12

TOPIC:-

VisionToll : AI-Powered Intelligent Vehicle Classification for Automated Toll Collection

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Abstract:

VisionToll is an intelligent transportation system that aims to automate the process of vehicle classification at toll plazas using deep learning and computer vision. The system analyzes images of incoming vehicles and accurately categorizes them into predefined classes such as cars, buses, trucks, motorcycles, and vans. Based on the identified vehicle category, the system can determine the corresponding toll category, reducing manual intervention, improving processing efficiency, and supporting faster, more reliable toll operations.

The proposed system will be implemented using a deep learning-based image classification model trained on publicly available vehicle image datasets. Vehicle images will undergo preprocessing before being used to train the model to recognize distinguishing visual features of different vehicle types. A simple user interface will allow users to upload a vehicle image and receive the predicted vehicle category, confidence score, and the applicable toll category. The project demonstrates the practical application of deep learning in intelligent transportation systems and highlights its potential for smart toll management and traffic analytics.